

## 🚧 RS003 Week 9 — English Worksheet

**Topic:** Keep the Aim Inside the Screen · **Course:** Pygame Turret · **Time:** about 45 minutes

This week you stop the barrel from flying off the screen. You set a **minimum** and **maximum** X with constants, then use comparisons ( `<` , `>` ) to keep the aim inside. The trick of forcing a value back inside a range is called **clamping**.

💬 Words to know: **minimum / maximum, compare ( `<` , `>` ), clamp, boundary, limit**

### 1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
MIN_X = 50
MAX_X = WIDTH - 50
if keys[pygame.K_LEFT] and gun_tip_x > MIN_X:
    gun_tip_x -= SPEED
```

The gun only moves left if two things are true. When does it stop moving left?

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```
if gun_tip_x < MIN_X:
    gun_tip_x = MIN_X
if gun_tip_x > MAX_X:
    gun_tip_x = MAX_X
```

This runs **AFTER** the gun moves. What does it do if the gun went too far left or too far right?

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```
pygame.draw.line(screen, RED, (MIN_X, 0), (MIN_X, HEIGHT), 2)
pygame.draw.line(screen, RED, (MAX_X, 0), (MAX_X, HEIGHT), 2)
```

**Two vertical red lines are drawn. What do they show the player?**

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## 2 · Run It

 **Type the example, run it, and hit the walls**

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))

MIN_X = 50
MAX_X = WIDTH - 50
gun_tip_x = CENTER_X
SPEED = 5
RED = (200, 0, 0)

clock = pygame.time.Clock()
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    if keys[pygame.K_LEFT] and gun_tip_x > MIN_X:
        gun_tip_x -= SPEED
    if keys[pygame.K_RIGHT] and gun_tip_x < MAX_X:
        gun_tip_x += SPEED

    screen.fill((135, 206, 235))
    pygame.draw.line(screen, RED, (MIN_X, 0), (MIN_X, HEIGHT), 2)
    pygame.draw.line(screen, RED, (MAX_X, 0), (MAX_X, HEIGHT), 2)
    pygame.draw.rect(screen, (0, 100, 0), (CENTER_X - 30, HEIGHT - 80, 60, 40))
    pygame.draw.line(screen, (80, 80, 80), (CENTER_X, HEIGHT - 80), (gun_tip_x,
HEIGHT - 200), 10)
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

**Aim all the way left, then all the way right. Where does the barrel stop? What stops it?**

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### 3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

**Bug A** — The gun should not move past the left wall, but it flies right off the left edge of the screen.

```
if keys[pygame.K_LEFT]:  
    gun_tip_x -= SPEED
```

**Hint:** there is no check that the gun is still past `MIN_X`.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug B** — The clamp is supposed to pull the gun back if it goes too far, but the comparisons are backwards so it teleports to the wrong wall.

```
if gun_tip_x > MIN_X:  
    gun_tip_x = MIN_X  
if gun_tip_x < MAX_X:  
    gun_tip_x = MAX_X
```

**Hint:** if the gun is **smaller** than `MIN_X` it went too far left. Check the directions.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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**Bug C** — The right boundary line is supposed to sit a little inside the right edge, but it is drawn off the screen and you cannot see it.

```
MAX_X = WIDTH + 50
```

**Hint:** the screen ends at **WIDTH** . To sit inside, the boundary must be a bit **less** than that.

**Write the fixed code:**

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**Why was it wrong? Why does your fix work?**

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## 4 · Modify It

### Make the play area narrower

Change `MIN_X` and `MAX_X` so the gun has a smaller space to aim in.

**Write the new values you chose:**

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### Try the clamp method instead of the guard method

Right now the example checks `> MIN_X` before moving. Rewrite it so the gun moves freely, then a clamp pulls it back inside afterwards.

**Write your clamp lines:**

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**Hint:** both methods keep the gun inside. The clamp lets it move first, then snaps it back if it went too far.

## 5 · Make It 📷

## Build a turret that stays on screen

Build a program where:

1. the barrel aims left and right,
2. it can never leave the screen,
3. red boundary lines show where the limits are.

Send a **photo or video** showing the gun stopping at both walls, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I set MIN\_X and MAX\_X to ...

The gun stops on the left because ...

The gun stops on the right because ...

## Clamping means ...

One tricky moment was when ...

If I had more time, I would ...



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## 6 · Record Your Walkthrough

Take a video while the gun hits both walls. Talk through it like you are teaching someone. Try to use these words: **minimum, maximum, compare, clamp, boundary.**

1. Aim into the left wall and show it stop.
2. Aim into the right wall and show it stop.
3. Read one boundary check out loud and explain it.
4. Say what would happen with no limits at all.

**Write what you will say in your video.** Plan it here before you record.

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**Submit** 

Send this worksheet + your photo or video to teacher on KakaoTalk.